

LAB EXPERIMENT RECORD

SET 9: ONLINE LEARNING ACTIVITY ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	Saturday

Aim

To analyze online learning activity data using a histogram, a boxplot, a bubble scatter plot and a smoothed line chart in R, and to build an interactive dashboard using Tableau.

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

PROGRAM 1: Histogram and Boxplot of Quiz Score

Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

```
# Import the dataset (keep the CSV in the working directory)
learning <- read.csv("Online_Learning_Activity.csv")

# read.csv() renames "Study_Time (hrs)" to "Study_Time..hrs."; rename it
names(learning)[names(learning) == "Study_Time..hrs."] <- "Study_Time"

# Check the imported data
head(learning)
summary(learning$Quiz_Score)
tapply(learning$Quiz_Score, learning$Course, mean)

# 1. Histogram of Quiz_Score
hist(learning$Quiz_Score,
     breaks = seq(60, 100, by = 10),
     col = "steelblue",
     border = "white",
     xlab = "Quiz Score",
     ylab = "Number of Students",
     main = "Histogram of Quiz Score")

# 2. Boxplot of Quiz_Score by Course
boxplot(Quiz_Score ~ Course, data = learning,
        col = c("lightblue", "lightgreen"),
        xlab = "Course",
        ylab = "Quiz Score",
        main = "Quiz Score by Course")
```

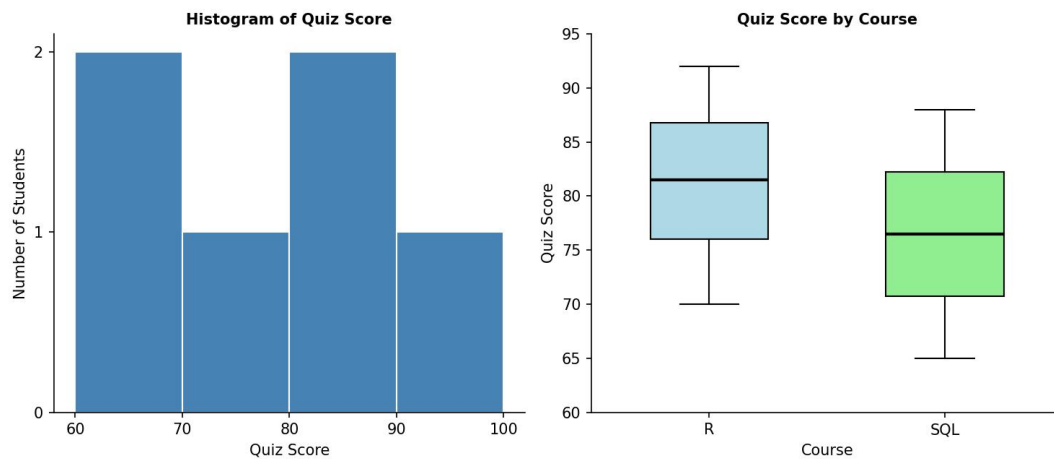
Console Output

```
Student_ID Gender Age Course Study_Time Videos_Watched Quiz_Score Login_Date
1          L01  Male  20      R          3.5             12         78 2025-01-05
2          L02 Female  22      R          4.2             15         85 2025-01-05
3          L03  Male  19    SQL          2.0              8         65 2025-02-08
4          L04 Female  21      R          5.0             18         92 2025-02-08
5          L05  Male  23      R          2.5              9         70 2025-03-12
6          L06 Female  20    SQL          4.0             14         88 2025-03-12

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
65.00  72.00   81.50   79.67  87.25   92.00

   R    SQL
81.25 76.50
```

Output 1a: Imported Data and Summary Statistics (R Console)



Output 1b: Histogram of Quiz Score and Boxplot of Quiz Score by Course

Interpretation: Quiz scores range from 65 to 92, with an average of about 79.7 and a median of 81.5. The histogram shows that the scores are spread over all four score ranges (2, 1, 2 and 1 students), so there is no single typical score and the performance of the class is mixed: three students scored 85 or more (85, 88 and 92), while two scored 70 or below (65 and 70).

In the boxplot, the R course has the higher median (81.5) than SQL (76.5) and the highest score (92), while the SQL scores are further apart (65 and 88). No outliers appear in either course. Since only six students are included (two in SQL), these differences should be treated as indications, not firm conclusions.

PROGRAM 2: Bubble Scatter Plot of Study Time vs Quiz Score

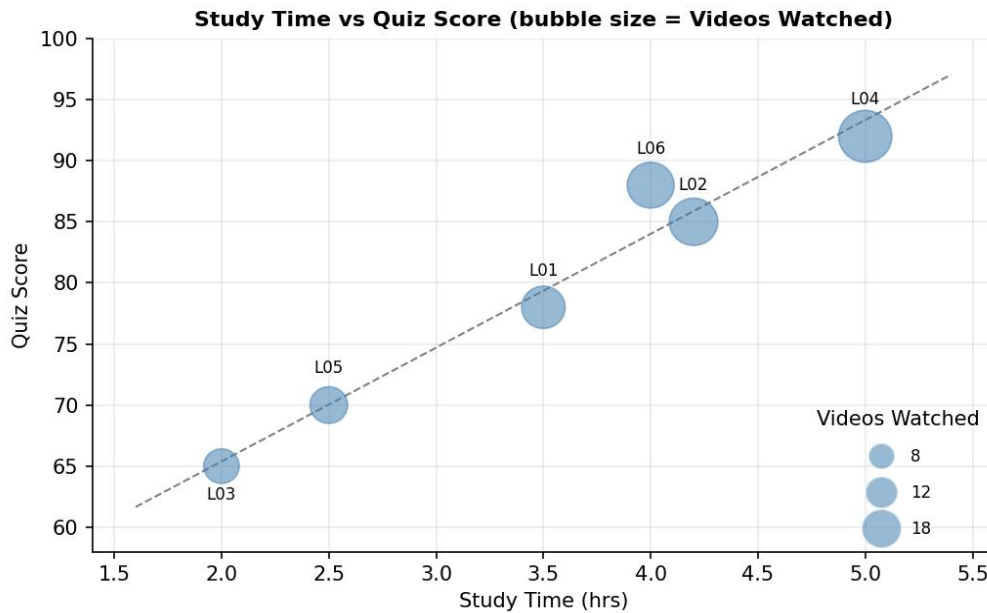
Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
library(ggplot2)

# Scatter plot: bubble size = Videos_Watched, transparency = alpha
ggplot(learning, aes(x = Study_Time, y = Quiz_Score, size = Videos_Watched)) +
  geom_point(colour = "steelblue", alpha = 0.55) +
  geom_text(aes(label = Student_ID), vjust = -1.8, size = 3) +
  geom_smooth(aes(x = Study_Time, y = Quiz_Score), method = "lm",
              se = FALSE, inherit.aes = FALSE,
              colour = "grey40", linetype = "dashed", linewidth = 0.6) +
  scale_size_area(max_size = 12, name = "Videos Watched") +
  labs(title = "Study Time vs Quiz Score (bubble size = Videos Watched)",
       x = "Study Time (hrs)", y = "Quiz Score")

# Strength of the relationships
cor(learning$Study_Time, learning$Quiz_Score)
cor(learning$Videos_Watched, learning$Quiz_Score)
```

OUTPUT:



Output 2: Study Time vs Quiz Score (Bubble Scatter Plot)

Learning pattern: The points rise from the lower left to the upper right, which shows a strong positive relationship: students who study longer score higher (correlation about 0.98). The fitted line suggests about 9 extra marks for each additional hour of study.

The bubbles also grow along the same direction, so students who watch more videos tend to score higher (correlation about 0.97). L04 studied the most (5.0 hours, 18 videos) and scored the highest (92), while L03 studied the least (2.0 hours, 8 videos) and scored the lowest (65). Transparency (alpha = 0.55) keeps overlapping bubbles such as L02 and L06 visible.

Note: Study time and videos watched are themselves almost perfectly related (correlation about 0.99), so their separate effects cannot be told apart. With six students, the pattern shows an association and does not prove that studying causes higher scores.

PROGRAM 3: Monthly Average Quiz Score with Moving Average

Using R programming, convert Login_Date into Date format. Calculate the average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
# Convert Login_Date to Date format
learning$Login_Date <- as.Date(learning$Login_Date, format = "%Y-%m-%d")
class(learning$Login_Date)

# Average quiz score per month
learning$Month <- format(learning$Login_Date, "%Y-%m")
monthly <- aggregate(Quiz_Score ~ Month, data = learning, FUN = mean)

# Moving average smoothing (window of 2 months)
monthly$Moving_Avg <- as.numeric(stats::filter(monthly$Quiz_Score,
                                              rep(1/2, 2), sides = 1))

print(monthly)

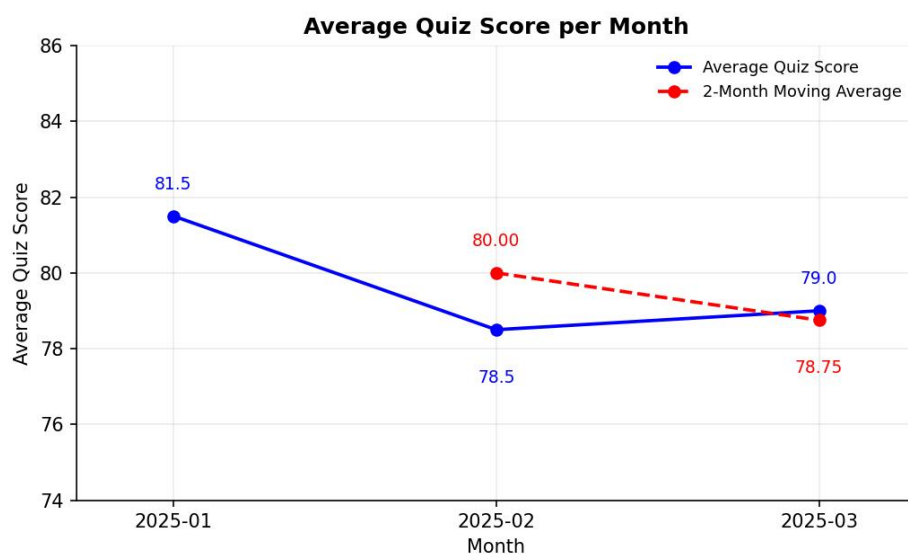
# Line chart with the moving average
plot(1:3, monthly$Quiz_Score,
     type = "o", col = "blue", lwd = 2, xaxt = "n", ylim = c(74, 86),
     xlab = "Month", ylab = "Average Quiz Score",
     main = "Average Quiz Score per Month")
axis(1, at = 1:3, labels = monthly$Month)
lines(1:3, monthly$Moving_Avg, type = "o", col = "red", lwd = 2, lty = 2)
legend("topright",
      legend = c("Average Quiz Score", "2-Month Moving Average"),
      col = c("blue", "red"), lty = c(1, 2), lwd = 2, pch = 1)
```

Console Output

```
[1] "Date"
```

	Month	Quiz_Score	Moving_Avg
1	2025-01	81.5	NA
2	2025-02	78.5	80.00
3	2025-03	79.0	78.75

Output 3a: Date Conversion and Monthly Averages (R Console)



Output 3b: Average Quiz Score per Month with Moving Average (Line Chart)

Trend: The average quiz score falls from 81.5 in January to 78.5 in February and then recovers slightly to 79.0 in March, a net change of -2.5 marks (about -3%). The 2-month moving average falls gently from 80.0 to 78.75, so the smoothed trend is slightly downward and then levels off, with the average staying within about 3 marks of 79 to 80. The overall performance is therefore fairly stable with a small decline after January.

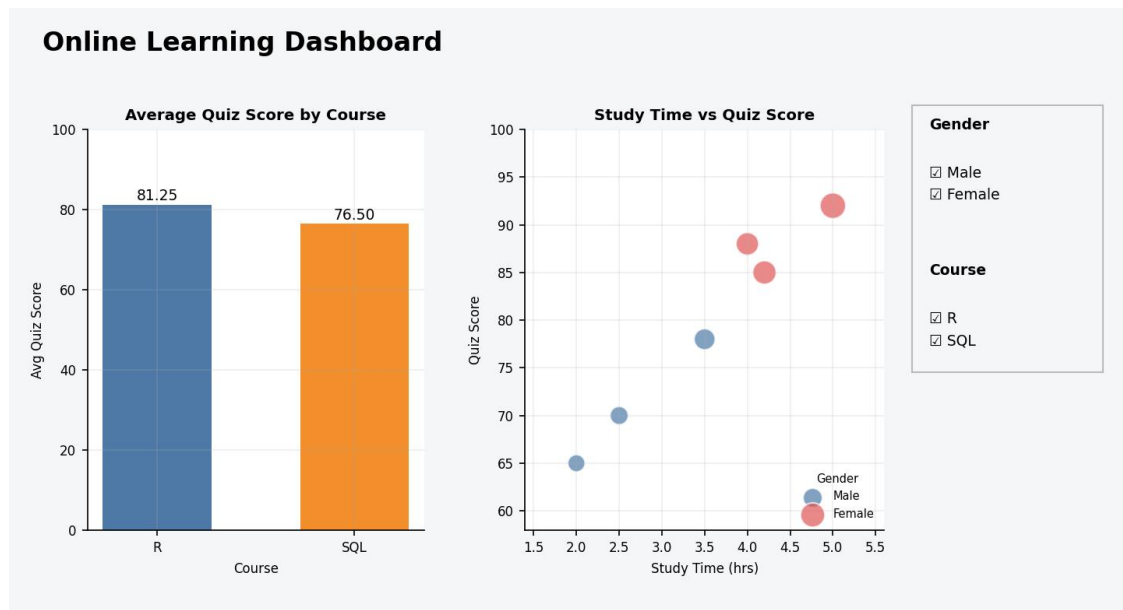
Note: There are only three months with two students in each, so the moving average has just two points and the trend is indicative only. The February average (78.5) hides a wide spread between L03 (65) and L04 (92).

PROGRAM 4: Interactive Learning Dashboard in Tableau

Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filters for Gender and Course. Design a mini interactive dashboard.

Steps in Tableau

1. Open Tableau and connect to the file using Connect > Text file > Online_Learning_Activity.csv. In the Data Source tab, check that Login_Date is a Date, Study_Time (hrs), Videos_Watched and Quiz_Score are numbers, and Student_ID, Gender and Course are strings.
2. Sheet 1 (Bar Chart): Drag Course to Columns and Quiz_Score to Rows, then right-click the Quiz_Score pill and choose Measure > Average. Drag Course to Color and AVG(Quiz_Score) to Label. Rename the sheet "Avg Score by Course".
3. Sheet 2 (Scatter Plot): Drag Study_Time (hrs) to Columns and Quiz_Score to Rows. Drag Student_ID to Detail so that every student appears as a separate point (or turn off Analysis > Aggregate Measures). Drag Videos_Watched to Size, Gender to Color, and Student_ID to Label. Rename the sheet "Study Time vs Score".
4. Add filters: drag Gender and Course to the Filters shelf, right-click each and choose Show Filter, and set Apply to Worksheets > All Using This Data Source so that both sheets are filtered together. Choose the Multiple Values (list) style.
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag both sheets onto the dashboard area. Place the Gender and Course filters in a container at the right side.
6. Add interactivity: select the bar chart and click Use as Filter, so clicking the R or SQL bar filters the scatter plot. Check that the tooltips show the Student_ID, Study Time, Quiz Score and Videos Watched.
7. Add the title "Online Learning Dashboard" and test the filters in Presentation Mode.



Output 4: Interactive Online Learning Dashboard (Tableau)

Observation: The bar chart shows that the average quiz score is higher in R (81.25) than in SQL (76.5), and the scatter plot shows that more study time goes with higher scores. The Gender and Course filters update both views at the same time. For example, selecting Female shows L02, L04 and L06 with scores of 85 to 92 (average about 88.3), while selecting Male shows L01, L03 and L05 (average about 71.0). The female students in this data also studied longer, so the difference should be read together with study time.

Procedure

1. Import the CSV file with `read.csv()`, rename the study time column, and check the data with `head()`, `summary()` and `tapply()`.
2. Use `hist()` to draw the histogram of Quiz_Score and `boxplot()` with a formula to draw the boxplot by Course.
3. Use `ggplot2` to draw the scatter plot with Videos_Watched as bubble size and alpha for transparency, and use `cor()` to measure the relationships.
4. Convert Login_Date with `as.Date()`, extract the month with `format()`, and calculate the monthly average with `aggregate()`.
5. Apply a 2-month moving average with `stats::filter()` and plot both lines with `plot()` and `lines()`.
6. Load the CSV into Tableau, build the bar chart and scatter plot, add the Gender and Course filters, and combine the sheets in an interactive dashboard.

RESULT:

The online learning activity data was successfully analyzed in R using a histogram, a boxplot, a bubble scatter plot and a monthly line chart with moving average smoothing. An interactive Tableau dashboard with Gender and Course filters was also created, so that users can explore quiz performance by course, gender and study time.

Key Functions and Tools Used

Function / Tool	Purpose
<code>read.csv()</code> / <code>names()</code>	Imports the CSV file and renames a column.
<code>hist()</code> / <code>boxplot()</code>	Creates the histogram and the boxplot by course.
<code>ggplot()</code> / <code>geom_point()</code>	Creates the bubble scatter plot with transparency.
<code>cor()</code>	Measures the relationship between two variables.
<code>as.Date()</code> / <code>format()</code>	Converts the login date and extracts the month.
<code>aggregate()</code>	Calculates the average quiz score per month.
<code>stats::filter()</code>	Applies moving average smoothing.
<code>plot()</code> / <code>lines()</code> / <code>legend()</code>	Draws the line chart with the moving average.
Tableau Filters	Filter the dashboard by Gender and Course.
Tableau Dashboard	Combines the bar chart and scatter plot with Use as Filter actions.